

Full-Stack AI Engineer



Ayoub Amazyan

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🌐 AmazyanAyoub 🌐 Portfolio

Summary

Full-Stack AI Engineer with three years of experience building production-grade AI systems across the full development lifecycle. Expertise spans agentic multi-agent systems, RAG pipelines, computer vision and OCR, model fine-tuning, conversational AI on Dialogflow CX, and end-to-end workflow automation. Proficient in Python, FastAPI, React, and AWS, with a strong track record of delivering complete solutions from data ingestion to deployed product across diverse industries.

Skills

Agentic AI & Automation : Multi-Agent Workflows (LangGraph, LangChain), Agentic AI Systems, n8n, Make, Relevance AI

Conversational AI : Dialogflow CX, NLU Engineering, Voice AI, Webhook Integration, Prompt Engineering

Computer Vision & OCR : OpenCV, Pillow, PaddleOCR, Vision-Language Models (LLaVA, Qwen2.5), NER (BERT)

RAG & LLM Frameworks : LlamaIndex, LangChain, Groq API, Ollama, Hugging Face, Qdrant, Pinecone, FAISS, ChromaDB

Model Training & Fine-Tuning : PyTorch, TensorFlow, HuggingFace, Custom Dataset Creation, ML-DL Pipelines

Programming : Python (primary), R, SQL, JavaScript

Deployment & Cloud : FastAPI, Docker, AWS (EC2, S3), DigitalOcean, Supabase, DVC

Data & Scraping : PostgreSQL, MongoDB, Neo4j, Playwright, Selenium

Experience

AI Agent Developer – Cnexia (Bell Canada Affiliate)

May 2026 – Present

- **Conversational Agent Development:** Design and build chat and voice agents on Dialogflow CX for Bell Canada's sales and customer service virtual assistants, implementing multi-turn dialogue flows, intent routing, and complex playbook architectures.
- **NLU Optimization & Prompt Engineering:** Diagnose and resolve NLU detection failures including intent misclassification, entity extraction errors, and training phrase coverage gaps. Apply advanced prompt engineering to improve LLM-driven routing accuracy.
- **Voice Agent Engineering:** Troubleshoot and optimize voice transcription pipelines, including speech recognition accuracy, SSML configuration, and telephony integration.
- **Webhook & Integration Debugging:** Identify and fix webhook failures, parameter passing errors, and fulfillment logic issues across multi-playbook agent architectures.
- **Technologies :** Dialogflow CX, Google Cloud, NLU, Voice AI, Webhook Integration, Prompt Engineering

Lead AI Automation Consultant – Independent

Jan 2024 – Present

- **Amazon Listing Automation Pipeline:** Built an end-to-end Python automation pipeline generating complete Amazon product listings across 11 international marketplaces (US, UK, DE, FR, IT, ES, CA, NL, SE, PL, BE) from a single source image. Auto-generates framed/unframed product mockups from PSD templates, produces AI-written titles, descriptions, and keywords via GPT-4o with automatic translation into 7 languages, uploads images to AWS S3, and writes fully-populated marketplace-compliant flat files including GPSR safety fields, localized sizing, and per-market regulatory data. Reduced a manual process taking hours per listing to a one-command workflow completing in minutes. **Tools:** Python, openpyxl, psd-tools, OpenAI API, boto3, AWS S3.
- **Autonomous Sales & Lead Qualification Agent:** Developed an end-to-end sales agent using Relevance AI that qualifies leads and handles FAQs. The agent autonomously retrieves lead details (name/organization), performs real-time web research to validate organization criteria, and makes go/no-go decisions. Upon qualification, it automatically schedules meetings via Calendly API and triggers Make webhooks to sync data into Pipedrive.
- **n8n Enterprise Automation Workflows:** Developed a full automation pipeline using n8n. Scraped structured content from public websites, cleaned and transformed the data, and embedded it into Supabase. Built a RAG-powered AI assistant using LangChain and Groq to answer queries with real-time accuracy.
- **Full-Stack AI Data Solutions:** Built a production-ready application featuring a ReactJS frontend and a LangGraph-powered FastAPI backend, enabling users to perform complex data transformations and visualizations via natural language.

- **Document Intelligence & RAG:** Developed RAG-powered assistants using LangChain, LlamaIndex, and Groq to enable real-time semantic search, CV ranking, and summarization across complex business reports.
- **Scalable Deployment:** Orchestrated and deployed full-stack AI services on DigitalOcean with persistent PostgreSQL storage and Clerk authentication.
- **Technologies :** Relevance AI, LangGraph, LangChain, n8n, Make, FastAPI, React, Supabase, Groq, LlamaIndex, DigitalOcean, AWS S3, OpenAI API, Pipedrive API

AI Engineer — Data Scientist – Ziwig, Rabat

2023 – May 2026

- **Corrective RAG Pipeline Optimization:** Developed a high-performance RAG system utilizing an 8B SLM to achieve high accuracy. Features include contextual enrichment, a hybrid indexer (dense and sparse), and Qdrant vector database storage. Implemented a dual-stage retrieval process using bi-encoders for search and cross-encoders for re-ranking with advanced prompt engineering.
- **Vision-Language OCR Systems:** Developed an advanced OCR pipeline leveraging Ollama-hosted vision-language models (LLaVA, Qwen2.5) and OpenCV to automate structured data extraction from identity documents, reducing manual workload by 70%.
- **Prompt Engineering:** Applied advanced prompt engineering to ensure clean, reliable JSON outputs from LLMs for automated identity verification systems.
- **Genomic Data Classification:** Built a DNA sequence classification model to detect endometriosis from FASTQ files, improving diagnostic accuracy by 20% through optimized preprocessing and hybrid ML-DL strategies.
- **Technologies :** Qdrant, Ollama (LLaVA, Qwen2.5), OpenCV, SLMs, Python, Prompt Engineering (JSON), ML-DL (Genomics)

Education

FST - Faculty of Sciences and Technology - Fez, Morocco <i>Master's in Data Science and Intelligent Systems</i>	<i>2021 – 2023</i>
FST - Faculty of Sciences and Technology - Fez, Morocco <i>Bachelor's in Computer Science</i>	<i>2018 – 2021</i>

Languages

- English (Fluent), French (Fluent), Arabic (Native).

Interests

- AI & Automation, Competitive Programming, Football, Bodybuilding.